

Seeding MCMC/HMC algorithms

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July 7, 2026

Abstract

The primary goal of this work is to illustrate the need of seeding MCMC/HMC algorithms when dealing with unobserved/latent variables.

1 Introduction

In real-world scenarios, researchers and practitioners often aim to explain the causes of a specific

2 Literature Review

[Maa et al. \(1996\)](#)

3 Methodology

In Statistics, without loss of generalization, we usually propose a stochastic mechanism to approximate the underlying law that is governing a phenomenon of interest, say $\pi(\mathbf{z}|\boldsymbol{\theta})$, where $\mathbf{z}$ is a set of attributes of interest and $\boldsymbol{\theta}$ is a random vector that fully defines such a stochastic mechanism.

Sometimes we only observed a subset of $\mathbf{z}$, that is, if $\mathbf{z} = (\mathbf{x}, \mathbf{y})$ we only have measurements for $\mathbf{y}$, say $\mathbf{y}^{obs}$.

We can simulate $\mathbf{x}$ using Algorithm [1](#).

Algorithm 1 Algorithm Name

Require: $\theta^{(t)}$ **Ensure:** $\theta^{(t)}, \mathbf{x}^{(t)}, \mathbf{y}^{(t)}$

- 1: $t = 0$
 - 2: **repeat**
 - 3: $t \leftarrow t + 1$
 - 4: $\theta^{(t)} \sim \pi(\theta)$
 - 5: $\mathbf{x}^{(t)}, \mathbf{y}^{(t)} \sim \pi(\mathbf{x}, \mathbf{y} | \theta)$
 - 6: **until** $d(\mathbf{y}^{(t)}, \mathbf{y}^{obs}) < d^{max}$
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4 Simulations

For a discrete-time setting, let us assume that a population experience an exponential growth with time-varying rate r_t , that is, $N_{t+1} = e^{r_t} N_t$ for $t = 1, 2, \dots, T - 1$. We can rewrite the time-varying rate as $r_t = f_t - m_t$, where $f_t, m_t \in \mathbb{R}^+$ are the fertility and mortality rates at time t , respectively. Now, at each time t , we can think of abundance before and after mortality, say N_t^{pre} and N_t^{post} , respectively. Thus, if M_t denotes the total mortality we have that $N_t^{post} = N_t^{pre} - M_t$.

Given the aforementioned population demographics, we can propose the following stochastic population mechanism:

$$\begin{aligned} M_t &\sim \text{Bin}(N_t^{pre}, 1 - e^{-m_t}) \\ N_{t+1}^{pre} &\sim \text{Poi}(e^{f_t} N_t^{post}) \end{aligned}$$

Furthermore,

$$\begin{aligned} \log(N_1^{pre}) &\sim \text{Poi}(e^{LN_0}) \\ \log(a_t) &\sim \text{N}(\mu_{la}, \sigma_{la}) \\ \log(b_t) &\sim \text{N}(\mu_{lb}, \sigma_{lb}) \\ \log(f_t) &\sim \text{N}(\mu_{lf}, \sigma_{lf}) \end{aligned}$$

with

$$\begin{aligned}\mu_{lm} &\sim N(m_{lm}, sd_{lm}) \\ \sigma_{lm} &\sim N(0, sd_{lm}) \\ \mu_{lf} &\sim N(m_{lf}, sd_{lf}) \\ \sigma_{lf} &\sim N(0, sd_{lf})\end{aligned}$$

5 Concluding Remarks

References

Maa, J.-F., Pearl, D. K., and Bartoszyński, R. (1996). Reducing multi-dimensional two-sample data to one-dimensional interpoint comparisons. *The Annals of Statistics*, 24(3):1069–1074.